Ship Detection in High-Resolution Optical Remote Sensing Images Aided by Saliency Information

Zhida Ren, Yongqiang Tang, Zewen He, Lei Tian, Yang Yang, and Wensheng Zhang

***Abstract*— Ship detection is a crucial but challenging task in optical remote sensing images. Recently, thanks to the emergence of deep neural networks (DNNs), significant progress has been made in ship detection. However, there are still two significant issues that must be addressed: 1) the high-resolution optical images may confuse the background with the ship, leading to more false alarms during detection and 2) the detector receives fewer positive samples due to the sparse and uneven distribution of ships in the optical remote sensing images. In this article, we innovatively propose using the saliency information to aid the ship detection task to tackle these two issues. To achieve this goal, we devise two novel modules, feature-enhanced structure (FES) and saliency prediction branch (SPB), to boost the capacity of ship detection in complex environments and propose a new sampling strategy named salient screening mechanism (SSM) to increase the number of positive samples. More specifically, SSM is adopted during the training phase to mine more positive samples from the ignored set. Then, in an end-to-end learning fashion, a neural network that incorporates our carefully designed FES and SPB is trained to gain more discriminative information for distinguishing the foreground and the background. To evaluate the effectiveness of our proposal, two new datasets HRSC-SO and DOTA-isaid-ship are constructed, which possess the annotation information for both object detection and saliency detection. We conduct extensive experiments on the constructed dataset, and the results demonstrate that our method outperforms the previous state-of-the-art approaches.**

***Index Terms*— Deep learning, high-resolution optical images, remote sensing, saliency detection, ship detection.**

I. Introduction

SHIP detection in optical remote sensing images has played an increasingly important role in the search and positioning of fishing vessels, environmental protection, and military surveillance [1]–[3]. This task aims to distinguish all the ships in the image and locate them using some forms, such as a horizontal box. The traditional methods [2], [4], [5] for ship detection generally use handcraft features to extract information from images to classify and locate ships. Such features must be carefully designed, and their generalization ability across different datasets may be poor.

In recent years, deep neural networks (DNNs) have advanced the development in the field of object detection. Till now, a huge number of DNN-based object detection algorithms have emerged, including Faster R-CNN [6], YOLO [7], SSD [8], and many others. Nevertheless, it may be infeasible to directly use them for the ship detection task due to some specific properties of the object in remote sensing images [9], [10]. Due to the special characteristics of ship detection, numerous DNN-based detection approaches have been proposed from different perspectives. These methods can be roughly classified into three categories: region-based methods [3], [11], [12], pixel-based methods [13]–[17], and the combination of region and pixel [18], [19]. The region-based methods aim to obtain the bounding box of the ship during detection. The pixel-based methods are designed to acquire a pixel-level ship segmentation map during detection, with saliency detection being the most common. In addition, some works have attempted to combine these two types of detection tasks. Nie *et al.* [19], for example, use a hyper-complex Fourier transform to extract saliency information from images, determine the region of interest (ROI), and then distinguish using a support vector machine. Xiao *et al.* [18] adopt a semantic segmentation map to generate the rotated ship candidate area.

Although the methods mentioned above have achieved promising performance for ship detection, there are still some challenges that need to be addressed. In this study, we mainly focus on the following two key issues.

1) While high-resolution optical remote sensing images contain a wealth of ship information, they also introduce a substantial amount of irrelevant, redundant, and even interfering noise. For instance, as shown in Fig. 1(a), the color of ships and the background will be similar due to the effects of lighting and material, reducing the contrast between the two. Besides, as shown in Fig. 1(b), some objects in the background of docked ships, such as piers, are similar in shape to the ships. These factors complicate the background and make it easily confused with the ships, resulting in an increased number of false alarms during detection.

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Zhida Ren, Zewen He, Lei Tian, and Wensheng Zhang are with the Research Center of Precision Sensing and Control, Institute of Automation, Chinese Academy of Sciences, Beijing 100190, China, and also with the School of Artificial Intelligence, University of Chinese Academy of Sciences, Beijing 101408, China (e-mail: renzhida2017@ia.ac.cn; hezewen2014@ia.ac.cn; tianlei2017@ia.ac.cn; zhangwenshengia@hotmail.com).

Yongqiang Tang and Yang Yang are with the Research Center of Precision Sensing and Control, Institute of Automation, Chinese Academy of Sciences, Beijing 100190, China (e-mail: yongqiang.tang@ia.ac.cn; yang.yang@ia.ac.cn).

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(a) (b) (c)

Fig. 1. Challenges in ship detection, including (a) color interference, (b) shape interference, and (c) fewer positive samples. In (a) and (c), red, green, yellow, and blue boxes stand for the ground-truth box, the positive sample, the negative sample, and the ignored sample, respectively. In (b), light blue box is the interfering target.

2) Ships are often sparsely and nonuniformly distributed in optical remote sensing images. An unsuitable sampling strategy may result in fewer positive samples being obtained by the detector, resulting in insufficient characterization of ships. Specifically, the dense sampling strategy adopted in RetinaNet [20] considers anchor boxes with intersection-over-union (IoU) values in the range of [0, 0.4] to be negative samples, those with IoU values more than 0.5 are considered positive samples, and those with IoU values in the range of [0.4, 0.5] are ignored. This sampling strategy tends to produce fewer positive with those "potential high-quality positive samples" [i.e., the blue boxes in Fig. 1(c)] being excluded from the network training.

In this research, we propose the corresponding solutions to the two problems described above. To tackle the first issue, we devise a simple and flexible network that incorporates the saliency information into the ship detection task, allowing the discriminative feature of the foreground in the optical image to be captured more effectively. Our network structure is based on RetinaNet, a classic one-stage detector. In addition, we propose two innovative structures: the feature-enhanced structure (FES) and the saliency prediction branch (SPB). FES is inspired by feature pyramid network (FPN) [21] and merges the feature maps once again in a top–down manner on the basis of FPN to reinforce supervised semantics. Moreover, FES further uses a skip connection to integrate the complementary information from different feature fusion stages. On the prediction branch, SPB devises a novel feature channel decreasing structure, which imports saliency information into the network in an end-to-end learning fashion and encourages the model to see more foreground. For the second issue, we propose a salient screening mechanism (SSM) to push the network to mine high-quality positive samples, including samples with a high IoU value and samples near actual objects. The saliency information is being used as a criterion in our SSM to select high-quality positive samples from the ignored set (i.e., anchor boxes with IoU locating within the range of [0.4, 0.5]). In such a way, the negative impact of a lack of sufficient positive samples can be mitigated to some extent. The main contributions of this article can be summarized as follows.

1) We propose a novel deep network model that devotes to exploiting the saliency information of optical images to assist ship detection task. In our proposal, an FES module and an SPB module are carefully designed to introduce saliency-aware information with more powerful foreground expression capabilities into the original ship detection task. Ship detection and saliency prediction are jointly optimized in a unified multi-task learning framework.
2) We put forward an SSM to increase the number of positive samples, which uses the saliency information as the criterion to select high-quality positive samples from the ignored set.
3) We construct two new high-resolution optical remote sensing ship datasets, called HRSC-SO and DOTA-isaid-ship, which contain both saliency and horizontal bounding box annotation of ships. The experimental results on two datasets confirm that our proposal is superior to multiple state-of-the-art methods.

We organize this article as follows: In Section II, related work is analyzed. Section III illustrates the details of our proposed method, including the basic framework, SSM, FES, prediction branch, and the multi-task loss function. Section IV conducts the experiments and ablation studies, followed by conclusion in Section V.

II. Related Work

Detecting objects in remote sensing images is a challenging complicated task. This article focuses on ship detection in high-resolution optical remote sensing images. In this part, we describe the recent related work in detail.

A. Object Detection

Given an image, object detection aims to determine where the objects are located and which class each object belongs to. Thanks to the emergence of DNNs, significant improvements have been achieved in object detection. As illustrated in [22], the existing DNN-based object detection methods for natural scenes can be grouped into two categories: two-stage networks and one-stage networks. The most well-known two-stage works are the R-CNN series, such as the Faster R-CNN [6], Cascade R-CNN [23], Libra Faster R-CNN [24], and many others. Although these methods achieve excellent accuracy, they face the challenges of high computational costs, time-consuming optimization processes, and massive computing power. In contrast, the one-stage approaches adopt a unified detection strategy, which directly feeds the image into a single network to predict the class of the target in the image and the bias of the bounding boxes. Representative methods include YOLO [7], SSD [8], RetinaNet [20] etc. In contrast to the two-stage methods, region proposal generation is no longer required in the one-stage models. Besides, the one-stage methods have a lower computational cost while maintaining comparable accuracy. In addition to the two categories mentioned above, it has been proven that instance segmentation [25], [26] can guide the model to locate the object more accurately in recent years. As a more complex task, instance segmentation can predict the target's bounding box and pixel-level semantic mask at the same time, which has attracted much attention.

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Although brilliant achievements have been achieved in object detection for natural images, it is not feasible to directly apply them to the ship detection task; when compared with natural images, ship detection in remote sensing images poses some specific properties [9], [10]. With the development of remote sensing technology, the resolution of optical remote sensing images is gradually increasing, making the targets easily confused with the background. For example, there are many ship-like targets (e.g., docks) in the inshore area, which affect the judgment of ships. In addition, the target's color, texture, and other features are difficult to distinguish from the background area.

B. Saliency Detection

In natural images, saliency detection aims to detect the salient objects in the image and segment the saliency targets in the way of pixel segmentation. In recent years, saliency detection has seen significant improvements in detection accuracy with the development of deep learning.

Saliency detection plays an important role in the field of computer vision, such as visual tracking [27] and robot navigation [28]. The traditional methods generally capture both local and global information, which are then merged to generate richer expressions. With the advent of convolutional neural networks, deep features can obtain rich ideographic information more quickly in virtue of the hierarchical expression capabilities of neural networks. In recent years, articles on saliency detection have paid more attention to the outline of the target. These articles [29]–[35] aim to obtain a more accurate target boundary using the hierarchical fusion of neural networks, the attention mechanism, and the loss of attention to the edge. Other articles [27], [36], [37] use salient object detection as some auxiliary tasks, including object segmentation [36], video object tracking [37], and many others. In addition, saliency detection [16], [38], [39] plays a very important role in the field of ship detection. Feng *et al.* [16] use the intensity difference and the phase spectrum of a Fourier transform to obtain the saliency map, which is then used to locate the candidate area in the panchromatic image. Dong *et al.* [38] partition the detection process into two stages: pre-screening and discrimination. In the pre-screening stage, the saliency of each pixel is judged using the variance of each region to guide positioning to the ROI. It can also suppress the majority of false alarms when obtaining a more precise position of the ship. Qi *et al.* [39] use the phase spectrum of Fourier transform (PFT) to generate a saliency map that guides the model in obtaining suspected ship targets with complete profiles.

Saliency detection occupies an important position in ship detection. However, most of the existing methods rely on handcraft features, resulting in low detection accuracy and generalization.

C. Ship Detection

Similar to object detection and saliency detection, ship detection can be divided into two categories according to the difference in feature extraction: traditional machine learning methods [2], [4], [5], [40], [41] and deep learning methods [3], [11], [12], [18], [42]–[48].

The traditional machine learning methods use handcraft features for feature extraction and adopt machine learning classifiers (e.g., SVM) to differentiate between the ship targets and the background. Zhu *et al.* [2] first extract the shape and texture features to capture the ROI. The semi-supervised multistage classifier is then used to classify the area of ships to eliminate false alarms. Bi *et al.* [4] propose to use the saliency information to search ROIs and design a neighborhood similarity-based method to describe local contextual information. Shi *et al.* [5] start by converting the panchromatic image into a "fake" hyperspectral image from which the ship candidate area can be obtained using a detector. Then, they add more discriminative features using the handcraft features in the candidate area. Han *et al.* [41] exploit the deep Boltzmann machine to learn the spatial and structural information from low-level features to obtain high-level representation. Meanwhile, many researchers begin to pay attention to deep learning methods due to their superior performance. In [3], a simple neural network is designed to obtain the ship candidate area and SVM is then used for classification. Lin *et al.* [11] devise a multi-scale convolutional structure to perform land–sea segmentation, which is used to obtain pixel-level ship detection area. Zhang *et al.* [12] design an anchor-free network for ship detection. Xiao *et al.* [18] use a semantic segmentation map to generate anchor boxes. Guo *et al.* [42] use Libra R-CNN to solve the imbalance problem in ship images. You *et al.* [43] combine convolutional neural network and saliency probability map for prow detection in optical remote sensing images. Chen *et al.* [44] propose a generalized Hough transform and a contour refinement to detect inshore ships. Ding *et al.* [45] propose an ROI transformer to detect densely packed objects in aerial images. Han *et al.* [46] incorporate rotation-equivariant networks into the detector to accurately predict the orientation of objects. Li *et al.* [47] propose a novel CNN-based algorithm to predict the orientation and other variables independently by creating two regression branches. Yang *et al.* [48] devise a rotation dense FPN to generate high-level semantic feature maps at all sizes, hence enhancing feature propagation. Although the current methods have achieved great performances with potentials, the performance of ship detection for high-resolution optical remote sensing images in the complicated background is still far from satisfactory.

III. METHODOLOGY

The overview of our method is shown in Fig. 2, a multitask pipeline based on multistage FPNs. We first use the SSM to uncover high-quality positive samples for training, and then we use a one-stage detector (i.e., RetinaNet) as the backbone to combine our proposed FES and SPB to obtain more discriminative information between foreground and background. The details of these parts are described below.

A. Network Structure of Ship Detection Aided by Saliency

The basic structure of our network uses the classic FPN structure [21] and we make several unique extensions to

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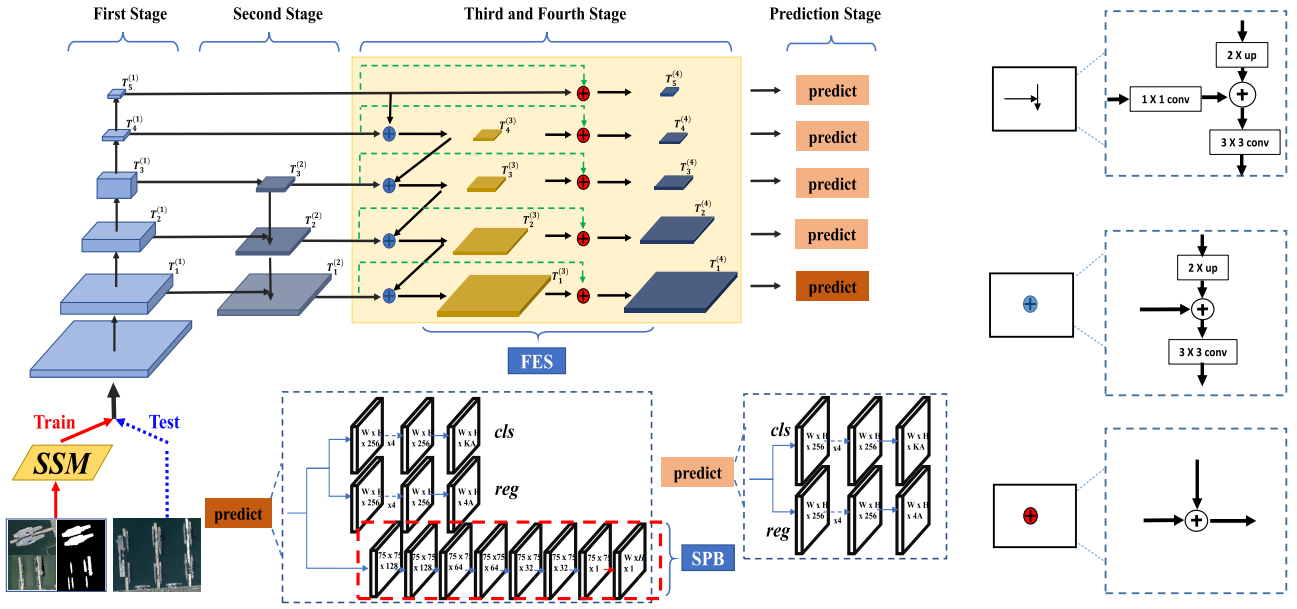


Fig. 2. Overview of our method. We introduce three additional parts to the basic RetinaNet. Specifically, we devise SSM to produce positive samples from the ignored set. It is worth noting that SSM is used only in the training phase. Besides, we add FES and SPB to distinguish the foreground and background for more effective ship detection. It should be noted that our prediction branch includes classification branch, regression branch, and SPB, which can borrow the advantages of multi-task learning to enhance performance.

the FPN structure for our purposes. We design the SSM to generate high-quality positive samples as the first extension. Specifically, SSM generates a saliency mask pyramid from the salient labels of the corresponding training images to obtain the saliency information in the position of the generated anchor boxes. In this case, IoU and saliency information can be coupled as a criterion to assign positive, ignored, or negative tags to the generated anchor boxes. In the second extension, we introduce the FES and SPB structures to distinguish foreground and background for more effective ship detection. More precisely, FES merges the hierarchical information in a novel top–down manner to enhance the expression ability of features. To reduce the risk of information loss and generate a saliency map of the input image, SPB fuses the multi-channel feature map with a new channel-decreasing structure.

To clarify, we introduce our network in detail by dividing it into four stages according to the number of feature pyramids generated during the forward propagation of networks. Formally, we use $T_i^{(j)}$ to denote the ith ($i \in [1, 5]$) level feature map of the feature pyramids generated in the jth ($j \in [1, 4]$) stage. The first stage network is a baseline network, which is used to extract features for the primary levels and generate the first feature pyramid, denoted as $T^{(1)} = \{T_1^{(1)}, T_2^{(1)}, T_3^{(1)}, T_4^{(1)}, \text{and } T_5^{(1)}\}$. The second stage network aims to merge the features from high to low and generate the feature pyramid $T^{(2)} = \{T_1^{(2)}, T_2^{(2)}, T_3^{(2)}, T_4^{(1)}, \text{and } T_5^{(1)}\}$. In the third and fourth stages, we use our proposed FES to generate their corresponding feature pyramids $T^{(3)} = \{T_1^{(3)}, T_2^{(3)}, T_3^{(3)}, T_4^{(3)}, T_5^{(1)}\}$ and $T^{(4)} = \{T_1^{(4)}, T_2^{(4)}, T_3^{(4)}, T_4^{(4)}, \text{and } T_5^{(4)}\}$, respectively. It should be noted that $T^{(4)}$ is exploited for object classification, bounding box regression, and saliency map generation.

B. Salient Screening Mechanism (SSM)

Generally, like RetinaNet [20], the anchor generation mechanism tries to produce K anchor boxes of different sizes at each corresponding grid position of the given image. Denote the anchor box set as A, and then each anchor box $a \in A$ will be assigned a label $k \in \{-1, 0, 1\}$. We partition all the anchor boxes into three sets based on their labels: positive set P (anchor boxes with label 1), negative set N (anchor boxes with label 0), and ignored set O (anchor boxes with label -1). Assume that $|A|$ is the number of elements in set A, and then we get

$$|P| + |O| + |N| = |A|. \tag{1}$$

To assign the label for each anchor box, people typically resort to the classical IoU metric, which is defined as

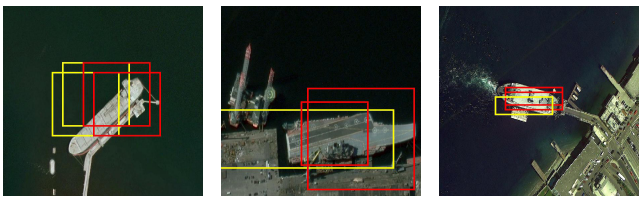
$$\text{IoU}(a, g) = \frac{a_s \cap g_s}{a_s + g_s - (a_s \cap g_s)} \tag{2}$$

where g_s represents the area of the ground-truth box g, and a_s is the area of the anchor box a. Then, we assign labels to the anchor boxes based on the following decision function:

$$k = \begin{cases} 1, & \text{IoU}(a, g) \geq 0.5 \\ 0, & \text{IoU}(a, g) \leq 0.4 \\ -1, & \text{otherwise.} \end{cases} \tag{3}$$

As shown in (3), the IoU-based sampling criterion adopted by RetinaNet rudely discards all the anchor boxes with IoU locating within the range of [0.4, 0.5], regardless of the fact that the samples within this range are not without merit. Actually, as shown in Fig. 3, some of the discarded samples (the red boxes) can be treated as positive samples, and the problem of lacking sufficient positive samples would be

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(a) (b) (c)

Fig. 3. Motivation of SSM. The IoU of all boxes displayed above locates within the range of [0.4, 0.5]. All the boxes are discarded according to the traditional IoU-based sampling criterion and the red boxes represent the positive samples recycled by our proposed SSM. (a) IoU: 0.41–0.42. (b) IoU: 0.45–0.46. (c) IoU: 0.49–0.50.

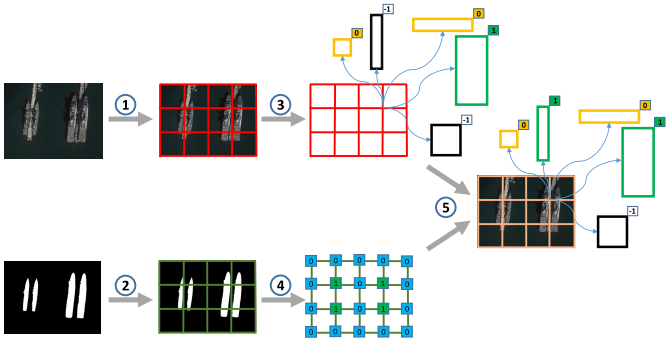


Fig. 4. SSM. The first step and the second step generate two kinds of grids, i.e., anchor generating grid and saliency grid, respectively. The third step produces K anchor boxes at every grid point and labels them according to the IoU metric, where 1, 0, and -1 represent the positive anchor box, negative anchor box, and ignored anchor box, respectively. The fourth step gets the label on the saliency grid point, where 1 and 0 denote the anchor box is salient or not. The fifth step matches the anchor boxes with the above two kinds of labels and relabels them.

alleviated to some extent if they could be effectively sorted out and recycled. To make use of these ignored samples, one intuitive strategy is using the IoU as an indicator to pick out the samples with high IoU value. Nevertheless, as exhibited by the yellow boxes in Fig. 3, the IoU criterion cannot guarantee to select appropriate positive samples. Some samples owning high IoU value but deviating from the main body of the ship object might affect the model training adversely. Inspired by this insight, we devise a new SSM to mine high-quality positive samples from the ignored set and use them to enrich the positive set. The details are illustrated in Fig. 4.

Specifically, as shown in Fig. 4, we also construct a saliency grid to obtain salient labels at the same position as the above grid when we construct the grid for anchor generation. After that, we reassign each anchor box a salient label $s \in \{0, 1\}$ according to the salient value, where 0 represents the anchor box is not salient and 1 represents the one that is salient. Thus, we get

$$s = \begin{cases} 1, & \text{if salient_value} > 0 \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

For an anchor box a, we have obtained k and s, and then a conversion mechanism is used to redistribute the final label z for a in the training phase. The conversion mechanism is

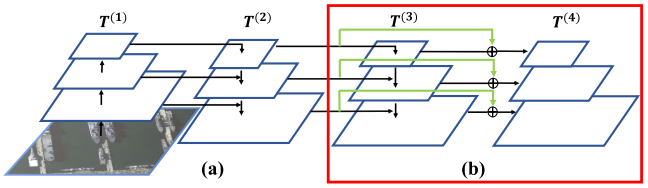


Fig. 5. Architecture of our FES. Based on the existing (a) FPN, our (b) FES further adds two-stage feature pyramids (i.e., $T^{(3)}$ and $T^{(4)}$). In the $T^{(3)}$ stage, the feature maps are fused in a top–down way once again cooperated with FPN's $T^{(2)}$ features, thus providing stronger semantic guidance for low-level features. In the $T^{(4)}$ stage, the complementary information of $T^{(2)}$ and $T^{(3)}$ is integrated using a skip connection shown by the green line.

formulated by (5). It should be noted that our SSM only works in the training phase of the model, which is shown in Fig. 2

$$z = \begin{cases} 1, & \text{if } k = 1 \text{ or } k = -1 \wedge s = 1 \\ 0, & \text{if } k = 0 \\ -1, & \text{if } k = -1 \wedge s = 0 \end{cases} \tag{5}$$

where 1, 0, and -1 represent that the anchor a is positive, negative, and ignored, respectively.

C. Feature-Enhanced Structure (FES)

The core idea of FPN [21] is to merge low-resolution semantically strong features into high-resolution semantically weak features via a top–down pathway and lateral connection [see Fig. 5(a)], such that a feature pyramid that has rich semantics at all the levels is built. In this study, we aim to introduce saliency information into ship detection while taking into account the complicated and ambiguous background. This means that compared with FPN where only bounding box annotation is available, additional supervised semantics are provided and deserve more thorough exploitation. To strengthen the expression ability of FPN, we propose the FES by combining hierarchical information more efficiently.

As illustrated in Fig. 5(b), our FES contains two-stage feature pyramids (i.e., $T^{(3)}$ and $T^{(4)}$ feature pyramids). In the $T^{(3)}$ stage, we fuse the feature maps in a top–down way once again cooperating with FPN's $T^{(2)}$ feature maps. In this manner, the rich semantic information can be further emphasized and provide stronger semantic guidance for low-level feature maps. In the actual operation of the $T^{(3)}$ stage, as illustrated in Fig. 2, we upsample the high-level features, add them directly to the low-level features, and then fuse them using a 3×3 convolutional layer like [29]. Despite reinforcing high-level semantics in the $T^{(3)}$ stage, the low-level texture information loss may occur due to the frequent use of upsampling, thus resulting in a low inter-class variance. For example, the shape of the dock is similar to that of a ship, making them easily misclassified into the same category. Carefully comparing $T^{(2)}$ with $T^{(3)}$, we can observe that in addition to the top–down semantic guidance, $T^{(2)}$ also contains the down–top subsampling hierarchical information provided by $T^{(1)}$, which are complementary to $T^{(3)}$. With this key insight, we integrate the complementary information of $T^{(2)}$ and $T^{(3)}$ using a skip connection [as shown by the green line

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in Fig. 5(b)]. The effectiveness of the two components above will be proven by the experiments in Section IV-E3.

D. Prediction Branch

As shown in Fig. 2, we make predictions on the prediction branch which consists of three child branches, i.e., classification branch, regression branch, and SPB. It should be emphasized that we conduct regression and classification of predicted boxes on all the features $T_i^{(4)}$ ($i \in [1, 5]$) and only predict the saliency map on $T_1^{(4)}$.

1) Classification Branch: This branch is used to obtain the class probability of the prediction box at each spatial location corresponding to each level feature map. We build this branch as [20], which includes four stacked sets of "3 × 3 Conv layer + ReLU," followed by a 3 × 3 Conv layer with KA channels. Finally, we use the sigmoid activation function to output the KA binary predictions per spatial location. The loss of this branch is the classical focal loss, which is designed to solve the class imbalance problem. The focal loss used in this article has the following formulation:

$$L_{\text{cls}} = -\alpha_t \log\left(p_t^*\right)/\gamma \tag{6}$$

where α_t represents the weight parameter of the positive and negative samples, and p_t^* is the classification probability of different classes, which is defined as follows:

$$p_t^* = \sigma(\gamma x_t + \beta) \tag{7}$$

$$x_t = yx \tag{8}$$

where $\sigma(\cdot)$ is the sigmoid function, x is the predicted class of a sample, γ and β are the hyperparameters $y \in \{\pm 1\}$ is the corresponding ground-truth class.

2) Regression Branch: Similar to the classification branch, the regression branch also contains four stacked sets of "3 × 3 Conv layer + ReLU," followed by a 3 × 3 Conv layer with $4A$ channels, which is used for predicting the parameterized coordinates $t_i = (t_x, t_y, t_w, t_h)$ of the predicted bounding box. The loss function of this branch is stated as follows:

$$L_{\text{reg}}\left(t_i, t_i^*\right) = \sum_{i \in \{x,y,w,h\}} p_i^* \text{smooth}_{L_1}\left(t_i - t_i^*\right) \tag{9}$$

in which

$$\text{smooth}_{L_1}(x) = \begin{cases} 0.5 \times x^2, & |x| < 1 \\ |x| - 0.5, & \text{otherwise} \end{cases} \tag{10}$$

$$\begin{aligned} t_x &= (x - x_b)/w_b, t_y = (y - y_b)/h_b \\ t_w &= \log(w/w_b), t_h = \log(h/h_b) \\ t_x^* &= \left(x^* - x_b\right)/w_b, t_y^* = \left(y^* - y_b\right)/h_b \\ t_w^* &= \log\left(w^*/w_b\right), t_h^* = \log\left(h^*/h_b\right) \end{aligned} \tag{11}$$

where x and y represent the center coordinates of the predicted box, and w and h are its width and height, respectively. Similarly, x_b, y_b, w_b, and h_b denote those of the anchor box b, while x^*, y^*, w^*, and h^* denote those of the ground-truth box. p_i^* is the ground-truth label that equals to 1 if the anchor box is positive and 0 if the anchor box is negative. $t_i^* = (t_x^*, t_y^*, t_w^*, t_h^*)$ represents the parameterized coordinates of the ground-truth box associated with a positive anchor box.

3) Saliency Prediction Branch: In our concerned ship detection task of remote sensing images, we design SPB, such as Mask R-CNN, to extract spatial information from the saliency mask to introduce ship saliency information into our model's training. The additional saliency mask requires the extraction of the much finer spatial layout of objects, which puts high demands on the SPB. Therefore, we select the features $T_1^{(4)}$ that contain more details to predict saliency map like [29], [49].

In Fig. 2, $T_1^{(4)}$ has a large number of feature channels and small feature sizes. Therefore, it is critical to minimize the risk of information loss while converting multichannel features into a single-channel saliency map. Otherwise, the auxiliary effect of the saliency branch on the entire network would most likely be seriously affected. To accomplish this, we create a new structure, SPB, which predicts the saliency map and allows saliency-aware information to be included into the features of each level. In Fig. 2, the SPB consists of three stacked sets of convolution structure, each of which contains two convolutional kernels with a size of 1 × 1 and 3 × 3, respectively. It is worth noting that the size of the feature channel varies and gradually decreases. Then, for fusion, we use a 1 × 1 Conv layer, followed by an interpolation algorithm to expand the single-channel feature to the original input size. Similar to Mask R-CNN, SPB predicts a binary class-agnostic foreground/background mask independently to decouple segmentation and classification. In particular, we use binary cross-entropy loss for this branch in the training phase, which is formulated as follows:

$$L_{\text{sal}} = -\sum_{i=1}^{m}\left[y_i \log\left(y_i^*\right) + (1 - y_i)\log\left(1 - y_i^*\right)\right] \tag{12}$$

where y_i presents the salient ground-truth value in pixel i, and y_i^* is the predicted salient value. m denotes the total pixels in the input image I.

E. Multi-Task Loss

Many previous articles [25], [26] have proven that multi-task learning can significantly enhance the performance of object detection. Inheriting this idea, we design a multi-task learning framework that combines the regression loss, classification loss of object detection, and salient object loss for end-to-end learning. Finally, we obtain the final loss function

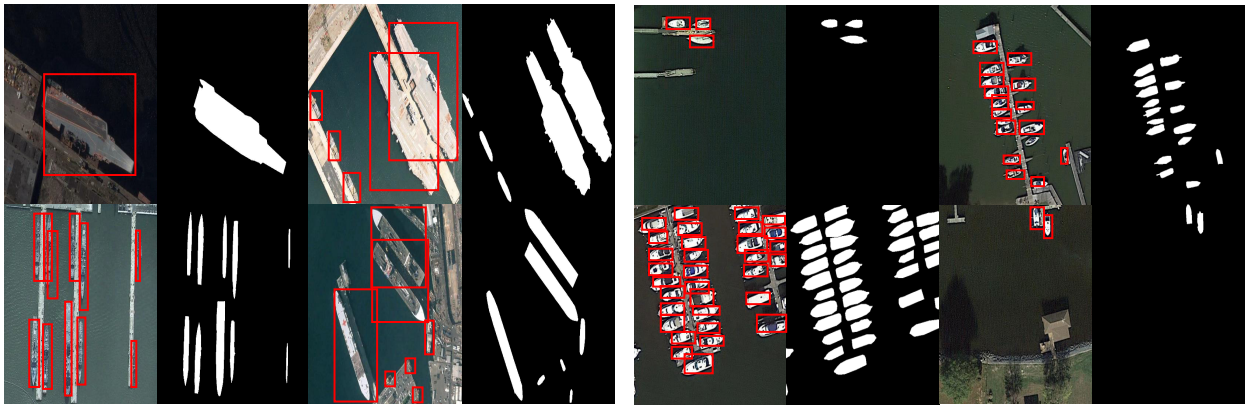
$$L = \frac{1}{N_{\text{reg}}} L_{\text{reg}} + \lambda_1 \frac{1}{N_{\text{cls}}} L_{\text{cls}} + \lambda_2 L_{\text{sal}} \tag{13}$$

where λ_1 and λ_2 are the hyperparameters used to balance the importance of the classification loss and the salient object loss, respectively. We use N_{reg} and N_{cls} to normalize regression loss and classification loss, respectively. In this article, both N_{reg} and N_{cls} represent the number of anchor boxes assigned to the ground truth.

IV. Experiments

In this section, we first introduce two rebuilt ship datasets, HRSC-SO and DOTA-isaid-ship. Then, the experimental setup

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(a) (b)

Fig. 6. Samples in (a) HRSC-SO and (b) DOTA-isaid-ship.

TABLE I
STATISTICS OF THE USED TWO DATASETS

Dataset	Type	Trainval set	Test set
HRSC2016	image object	436 1,207	453 1,188
HRSC-SO	image object	429 1,165	626 1,719
DOTA	image object	354 33,307	301 12,453
DOTA-isaid-ship	image object	3,854 61,001	4,709 21,758
DOTA-isaid-ship-R	image object	3,732 58,483	4,709 21,758

is described in detail, followed by the comparison between our method and several state-of-the-art methods. Finally, the analytical experiments and visual results are presented to verify the effectiveness of each part in our method.

A. Dataset and Analysis

1) Dataset: Our approach requires extra salient labels, and thus existing datasets are generally unsuitable for evaluation purposes. To remedy this issue, we construct two new high-resolution ship datasets in the optical satellite image, called HRSC-SO and DOTA-isaid-ship. Table I shows the statistics for the two datasets.

HRSC-SO is generated from HRSC2016 [50], with samples shown in Fig. 6(a) and our trainval set contains 429 images derived from the HRSC2016 semantic segmentation set. Moreover, we transform the original semantic labels into salient labels and select the samples with ship objects in HRSC2016 as the test set of HRSC-SO, which contains 626 images without salient labels.

DOTA-isaid-ship is generated from DOTA-v1.5[1] and iSAID [51]. As a large-scale aerial image dataset, DOTA-v1.5 contains a large number of remote sensing targets. The iSAID dataset is a newly developed large-scale instance segmentation dataset that uses image data from the DOTA-v1.5 dataset. To validate our experiment, we obtain the image and position annotation information of the ship from the train set and eval set of DOTA-v1.5. Meanwhile, the saliency mask of the ship is extracted from the corresponding image in the iSAID. Due to the big resolution of DOTA-v1.5 ranging from 800–13 000, we cut all the collected images into 600 × 600 using the processing method provided by the open tool DOTA_devkit[2] for the sake of reducing memory consumption. Finally, we crop out 8827 images on the training set and 4709 images on the test set. Furthermore, to facilitate the comparison of different methods designed for horizontal or oriented bounding box tasks and produce two versions of the dataset, namely, DOTA-isaid-ship [horizontal boxes, see Fig. 6(b)] and DOTA-isaid-ship-R (oriented boxes). For these two datasets, we adopt DOTA_devkit to eliminate the invalid training images. In the end, on the trainval set, DOTA-isaid-ship contains 3854 images and 61 001 ships. Meanwhile, DOTA-isaid-ship-R has 3731 images and 58 483 ships.

It should be noted that the saliency mask in two datasets uses the annotation rules from Pascal-S [52] and the form is based on pixel-level labeling like DUT-OMRON [53], where the label value is 0 or 255. To be more specific, 255 denotes the salient position associated with a ship, while 0 denotes the background, i.e., the remaining positions without ships.

2) Dataset Analysis: To gain a deeper understanding of new datasets, we conduct a more comprehensive analysis for HRSC-SO and DOTA-isaid-ship, as shown in Figs. 7 and 8, respectively. Besides, more details about HRSC-SO can be found in Appendix.

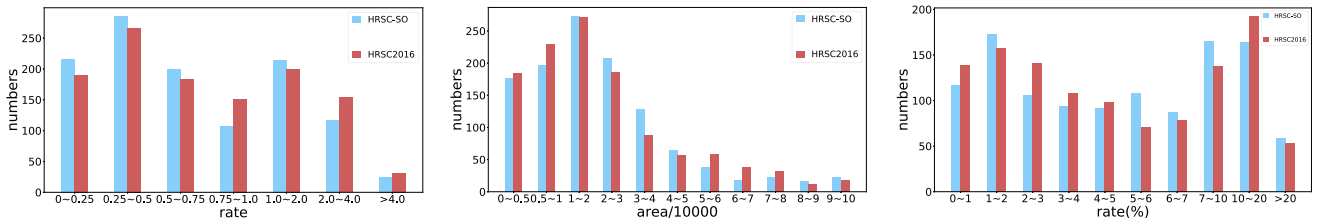
a) Comparison between HRSC-SO and HRSC2016:

1) *The Amount of Samples:* Table I shows very few differences in the number of two sets regardless of the image or object. However, the size of our test set is larger than that of HRSC2016, which contains samples of three sets in HRSC2016 (see Appendix).
2) *The Distribution of Ship Aspect Ratio:* Fig. 7(a) depicts the distributions of the ship aspect ratio and it is obvious that the two distribution are relatively similar.

[1]https://captain-whu.github.io/DOAI2019/challenge.html

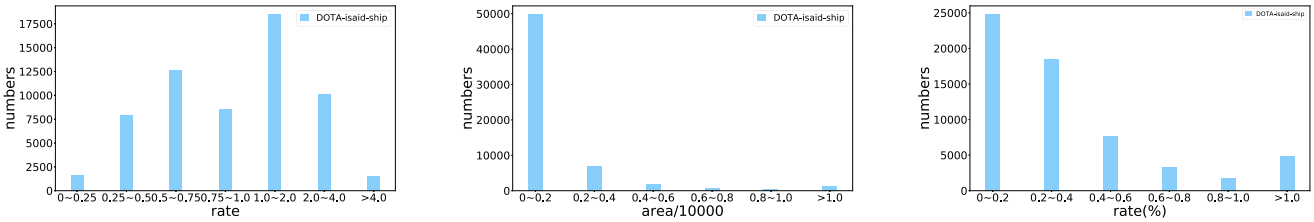
[2]https://github.com/CAPTAIN-WHU/DOTA_devkit

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(a) (b) (c)

Fig. 7. Comparisons between the trainval set of HRSC-SO and the training set of HRSC2016. (a) Ship aspect ratio distribution. (b) Ship area distribution. (c) Ship area ratio distribution.



(a) (b) (c)

Fig. 8. Analysis of the trainvel set in DOTA-isaid-ship. (a) Ship aspect ratio distribution. (b) Ship area distribution. (c) Ship area ratio distribution.

Nevertheless, our training set has more samples with an aspect ratio of less than 0.25, indicating that the training set of HRSC-SO has more "narrow and long" targets.

3) *The Distribution of Ship Area:* The ship area distribution is displayed in Fig. 7(b) and (c). We can observe that the area distribution is similar and uneven between the two training sets. In addition, as shown in Fig. 7(c), two training sets contain a large number of ships with a small area.

b) DOTA-isaid-ship: By comparing the statistics of HRSC-SO with DOTA-isaid-ship in Table I, some differences are revealed. First, DOTA-isaid-ship contains more instances. In addition, the aspect ratio is more concentrated around 1.0 [see Fig. 8(a)] and DOTA-isaid-ship includes more small ships, as shown in Fig. 8(b) and (c).

B. Experimental Setup

1) Experimental Setting: We use the PyTorch framework and implement all the experiments on a standard Linux server with CUDA v10.0, cuDNN v7.6.0, and TITAN Xp with 12-GB GPU memory. We use the SGD optimizer. The batch size is set to four for HRSC-SO and eight for DOTA-isaid-ship. In the training phase, we train 800 epochs with a learning rate of 0.001, a weight decay of 0.0001, and of momentum 0.9. We apply NMS with a threshold of 0.5 to produce the final result. Besides, in our experiments, we fix $\lambda_1 = 1.0$ and $\lambda_2 = 0.000005$ for all cases. The parameters in the focal loss, i.e., α_t and γ, are set to 0.35 and 2, respectively.

2) Competitors: To demonstrate the effectiveness of our proposed method, we compare it with ten representative methods equipped with different backbones, which can be divided into three categories. The first category contains several popular general object detection methods, Faster R-CNN [6], Cascade R-CNN [23], Libra Faster R-CNN [24] and SSD512 [8].

TABLE II
AP WITH DIFFERENT ANCHOR BOX RATIOS

Aspect ratios	AP (%)
{1/2, 1, 2}	69.13
{1/4, 1, 3}	**85.52**
{1/3, 1, 2}	82.54
{1/4, 1, 2}	85.19
{1/3, 1, 3}	83.82

The second category is the remote sensing object detection methods based on instance segmentation, including widely used Mask R-CNN [25], Cascade Mask R-CNN, and hybrid task cascade (HTC) [26], which are provided in DOTA's benchmark for experimental comparison [46]. The third category falls into object detection in remote sensing images, including baseline methods (i.e., RetinaNet), Faster ROI transformer [45], ReDet [46], ANCB [47], and R-DFPN [48], where ANCB and R-DFPN are designed for ship detection task.

3) Evaluation Metric: For fairness, we adopt the average precision (AP) as the evaluation index, which is calculated as follows:

$$\mathrm{AP} = \int_0^1 P(R) dR \tag{14}$$

where P represents the precision rate and R is the recall rate. However, as illustrated in [43], the AP value will be influenced by the IoU threshold. Therefore, in the subsequent experiments, the AP results are produced with IoU $= 0.5$ if no special instructions are given. Furthermore, we also conduct experiments to demonstrate the evolution of AP as the threshold of IoU varies.

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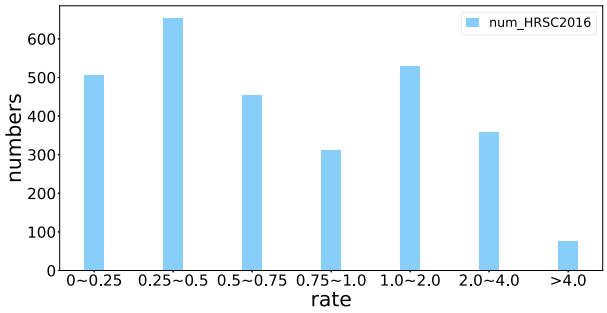


Fig. 9. Aspect ratio distribution on the HRSC-SO dataset.

C. Anchor Box Analysis

The aspect ratios of anchor boxes are universal in natural image datasets such as ImageNet [54] and MS-COCO [55]. However, as presented in [9], there exists a discrepancy between remote sensing images and natural images in terms of the imaging system and target state. As a result, the aspect ratios used in natural images may not be applicable to remote sensing images if the model is not robust to the changes in the size of the preset anchor box. In light of the above case, we run additional experiments to observe how the pre-defined anchor boxes affect our basic RetinaNet model. We first calculate the aspect ratios of the HRSC-SO dataset and report the results in Fig. 9. It is shown that the distribution of target size is uneven, and thus the traditional predefined anchor boxes are not suitable. To find a better anchor box ratio, we conduct experiments with various anchor box ratios and summarize the results in Table II.

The model achieves the highest AP score when the anchor box ratio is $\{1/4, 1, 3\}$. Therefore, we set the anchor box ratio to be $\{1/4, 1, 3\}$ for the HRSC-SO dataset in the following experiments. Besides, we can find that the universal anchor box ratio $\{1/2, 1, 2\}$ used in natural image datasets obtains a very low AP score on HRSC-SO, which falls behind the highest one by 16.39%. This phenomenon demonstrates the importance of determining an appropriate anchor box ratio for remote sensing images in ship detection.

The anchor aspect ratio in the DOTA-isaid-ship dataset is concentrated around 1.0 as shown in Fig. 8(b). Hence, in this study, we use the default preset value of the RetinaNet model, i.e., $\{1/2, 1, 2/1\}$, for the DOTA-isaid-ship dataset.

D. Comparison to the State-of-the-Art Methods

Our model is evaluated on two datasets and compared with ten excellent approaches.

1) Experimental Results on HRSC-SO Dataset: The comparison results of our method and all competitors with various backbone networks are shown in Table III, where the best performance is boldfaced. As we can see, on the premise of outstripping baseline (RetinaNet), our approach outperforms all comparison methods regardless of the backbone network. To be specific, compared with the methods using Vgg16 backbone, i.e., Faster R-CNN and SSD512, our method increases by 13.54% and 2.4%, respectively. When the Resnet50 backbone is used, our proposed method outperforms general object detection methods, i.e., Faster R-CNN, Libra Faster R-CNN, Cascade R-CNN, and SSD512 by 13.46%, 5.3%, 5.0%, and 2.4%, respectively. We also make a comparison with three instance segmentation methods, i.e., Mask R-CNN, Cascade Mask R-CNN, and HTC. The results indicate that our proposal is superior to all of them. Moreover, it should be mentioned that our method also beats Faster ROI transformer and ReDet, both of which are designed specifically for remote sensing images. Finally, our method performs the best using the Resnet101 backbone, with an AP score of 89.69%. It is also worth noting that when compared with the other two ship detection methods, ours is significantly higher in terms of AP. The above results verify the significant superiority of our method against the state-of-the-art competitors.

TABLE III
AP OF DIFFERENT METHODS WITH VARIOUS BACKBONES ON THE HRSC-SO DATASET

Method	Backbone	AP(%)
Faster R-CNN [6]	Vgg16	75.56
	Resnet50	75.64
	Resnet101	79.54
Cascade R-CNN [23]	Resnet50	84.10
	Resnet101	87.74
Libra Faster R-CNN [24]	Resnet50	83.80
	Resnet101	84.58
SSD512 [8]	Vgg16	86.70
Mask R-CNN [25]	Resnet50	86.48
	Resnet101	87.82
Cascade Mask R-CNN	Resnet50	88.15
HTC [26]	Resnet50	86.59
	Resnet101	88.83
Faster ROI transformer [45]	Resnet50	85.10
	Resnet101	86.91
ReDet [46]	Resnet50	89.03
ANCB [47]	Vgg16	80.30
R-DFPN [48]	Resnet50	79.25
	Resnet101	80.14
RetinaNet [20]	Resnet50	85.52
	Resnet101	85.94
Ours	Resnet50	**89.10**
	Resnet101	**89.69**

2) Experimental Results on DOTA-Isaid-Ship Dataset: We also carry out experiments on the DOTA-isaid-ship dataset, and the results are presented in Table IV. We achieve 81.03% AP with the Resnet50 backbone. Compared with the general object detection methods, our model clearly surpasses Faster R-CNN (Resnet50), Cascade R-CNN (Resnet50), Libra Faster R-CNN (Resnet50), and SSD512 (Vgg16). With the Resnet101 backbone, we achieve 81.41% AP, leading Faster R-CNN (Resnet101), Cascade R-CNN (Resnet101), and Libra Faster R-CNN (Resnet101) by 4.41%, 4.41%, and 4.81%, respectively. Compared with the instance segmentation methods including Mask R-CNN, Cascade Mask R-CNN, and HTC, our proposal equipped with Resnet50 backbone achieves remarkable improvements over them, leading their corresponding backbone models by 23.32%, 10.72%, and 19.39%, respectively. In comparison to the state-of-the-art methods that are designed for remote sensing images, our proposed approach outperforms Faster ROI transformer, ReDet,

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TABLE IV
AP OF DIFFERENT METHODS WITH VARIOUS BACKBONES ON THE DOTA-ISAID-SHIP DATASET

Method	Backbone	AP(%)
Faster R-CNN [6]	Resnet50	75.90
	Resnet101	77.00
Cascade R-CNN [23]	Resnet50	75.80
	Resnet101	77.00
Libra Faster R-CNN [24]	Resnet50	73.90
	Resnet101	76.60
SSD512 [8]	Vgg16	62.60
Mask R-CNN [25]	Resnet50	57.71
	Resnet101	69.46
Cascade Mask R-CNN	Resnet50	70.31
HTC [26]	Resnet50	61.64
	Resnet101	70.01
Faster ROI transformer [45]	Resnet50	59.42
	Resnet101	70.06
ReDet [46]	Resnet50	70.59
	Resnet50	76.47
R-DFPN [48]	Resnet101	78.32
RetinaNet [20]	Resnet50	79.26
	Resnet101	79.70
Ours	Resnet50	**81.03**
	Resnet101	**81.41**

and R-DFPN by a large margin based on the backbone of Resnet50. Similar to the HRSC-SO dataset, our proposal obtains the optimal detection results with 81.41% AP on the DOTA-isaid-ship dataset when the Resnet101 backbone is used. In summary, the well-performed experimental results on DOTA-isaid-ship confirm that our model owns excellent generalization ability on the more challenging dataset.

E. Analytical Experiments

In this section, we conduct several analytical experiments on HRSC-SO and DOTA-isaid-ship to evaluate the effectiveness of our proposed method. Firstly, we carry out the ablation study to analyze the contributions of our proposed three components, i.e., SSM, FES, and SPB, respectively. We summarize the AP scores of our method with different components in Table V. Second, we also investigate the impact of FES and SPB structure. The results are shown in Tables VI and VII, respectively. Based on the experimental results, our method will be analyzed in more detail as follows.

1) Effectiveness of SSM: As described in Section III-B, SSM combines IoU value and the saliency information as the criterion for mining positive samples in the training phase. In such a case, we may be able to increase the number of positive samples, making it easier to train a better model. As shown in Table V, compared with RetinaNet, RetinaNet + SSM achieves an improvement of 1.11% in HRSC-SO. Similarly, the baseline coupled with the new sampling method SSM improves the performance of the original model by 0.55% on DOTA-isaid-ship. To further verify its effectiveness, we vary the IoU thresholds and report the AP results in Fig. 10. The results reveal that RetinaNet + SSM consistently performs

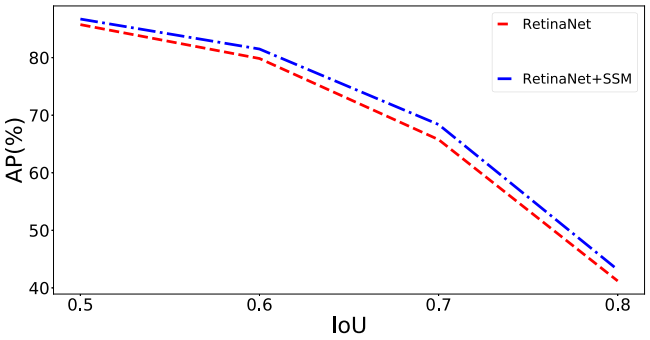


Fig. 10. AP of RetinaNet and RetinaNet + SSM with varying IoU thresholds. It can be found that RetinaNet using SSM can obtain better detection performance.

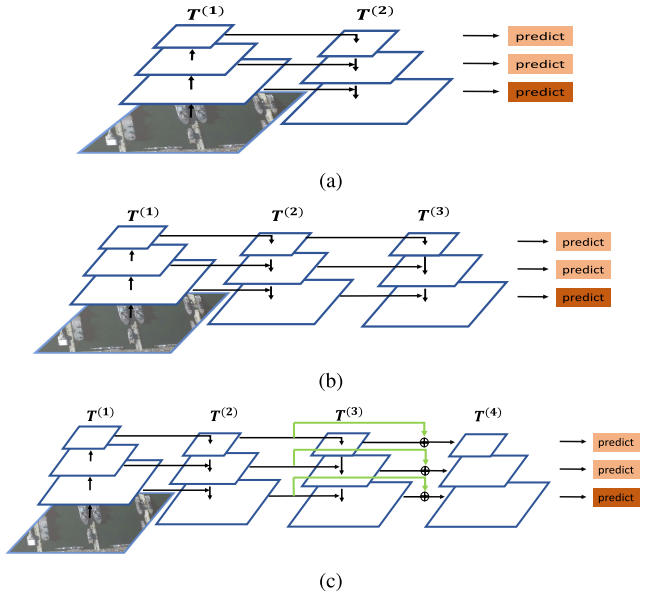


Fig. 11. Exploration of FES. (a) C1 and (c) C3 represent FPN and FPN + FES, respectively. (b) C2 is the structure of fusion in a top–down manner, which is based on C1. $T^{(i)}$ represents the feature pyramids of different stages. Our prediction head contains two structures. Light brown prediction head includes classification branch and regression branch, while dark brown prediction head adds a saliency prediction on this basis.

better than RetinaNet, indicating that our SSM possesses outstanding ability to mine more high-quality positive samples, promoting the detector to perform better regardless of IoU thresholds.

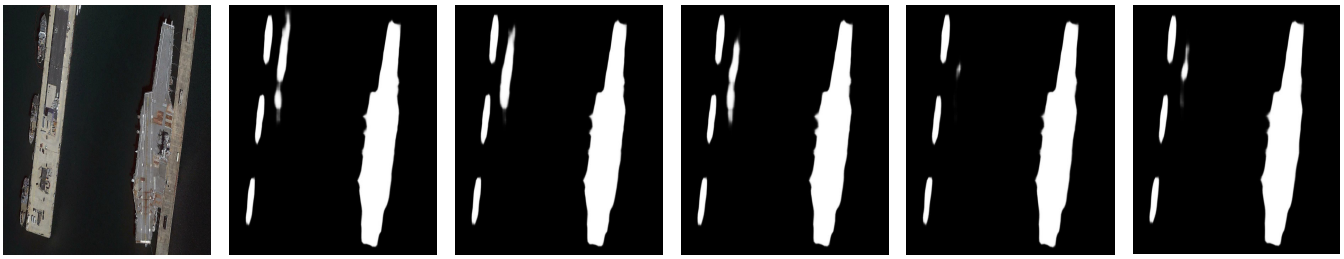
2) Effectiveness of FES and SPB: For the sake of obtaining more discriminative information between foreground and background, we propose a novel network equipped with FES and SPB. As we can see from Table V, the effect of the model is improved whether two parts are added separately or together. Specifically, on HRSC-SO, RetinaNet + FES and RetinaNet + SPB perform better than RetinaNet by 1.49% and 0.99% in AP, respectively. Furthermore, we can obtain a huge improvement up to 2.82% if we add both the parts into the basic RetinaNet, which indicates that joining the two parts can produce discriminative features more effectively than exploiting just one part. Besides, we can draw similar conclusions via the ablation study on the DOTA-isaid-ship

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(a) (b) (c)

Fig. 12. Visualization of FES with different structures. The corresponding detection results of the three models, i.e., (a) C1, (b) C2, and (c) C3, are shown above. Compared with C1, semantic-information-enhanced structure (C2) can better improve detection accuracy (see the results in the second column). Furthermore, by introducing a skip connection structure, our model (C3) avoids detecting some targets that are easily confused with ships, such as docks.



(a) (b) (c) (d) (e) (f)

Fig. 13. Predicted saliency mask of different structure. S1–S5 represent the saliency mask predicted from SPB with different structures which can prove that our structure (S4) can obtain better predicted performance. (a) Image. (b) S1. (c) S2. (d) S3. (e) S4. (f) S5.

TABLE V
PERFORMANCE COMPARISON OF DIFFERENT COMPONENTS ON TWO DATASETS

Dataset	Method	AP(%)
HRSC-SO	RetinaNet	85.52
	RetinaNet+SSM	86.63
	RetinaNet+FES	87.01
	RetinaNet+SPB	86.51
	RetinaNet+FES+SPB	88.34
	ours	**89.10**
DOTA-isaid-ship	RetinaNet	79.26
	RetinaNet+SSM	79.81
	RetinaNet+FES	80.11
	RetinaNet+SPB	79.89
	RetinaNet+FES+SPB	80.34
	ours	**81.03**

dataset. For FES, we observe a 0.85% increase. This indicates that the discriminative capability of the detector is improved by enhanced features. By adding SPB, the network obtains more spatial information in the images and achieves 79.89% AP. Eventually, a 1.08% improvement in AP is obtained by jointly using FES and SPB.

3) Structure Exploration of FES: In Fig. 11, we change the structure of FES while keeping SPB and SSM fixed to explore how effective two main components in our FES are. Compared with the C1 model, the C2 model has an additional feature fusion process via a top–down pathway and lateral connection. The C3 (i.e., our FES) model further adds a skip connection on the basis of C2 model. Through the experimental results shown in Table VI, we can observe that our FES (C3) performs better than other models. Specifically, in terms of AP, the C2 model is 1.08% higher than the C1 model and the C3 model further achieves 1.06% improvement on top of the C2 model. This phenomenon confirms the effectiveness of the added feature fusion structure and the skip connection structure in our FES.

TABLE VI
EXPERIMENTAL RESULTS WITH DIFFERENT STRUCTURE OF FES

Model	AP(%)
C1	86.96
C2	88.04
C3	**89.10**

To obtain a more intuitive understanding of FES, we conduct visual experiments on the above three models. As shown in Fig. 12, C2 has a higher detection accuracy [see Fig. 12(b)] than C1 by means of enhanced semantic information fusion. In addition, by introducing skip connections, C3 integrates more fine-grained feature information from the previous stage, potentially improving the inter-class variance. Objects in the background that are easily confused with ships, such as docks, can be eliminated in this manner.

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TABLE VII
EXPERIMENTAL RESULTS WITH DIFFERENT STRUCTURES OF SPB

Name	Structure	AP (%)
S1	256-1	87.95
S2	256-128-1	88.25
S3	256-128-64-1	88.23
S4	256-128-64-32-1	**88.34**
S5	256-128-64-32-16-1	87.95

4) Structure Exploration of SPB: As described in Section III-D, SPB is a channel-decreasing structure, which aims to make saliency prediction on $T_1^{(4)}$. SPB includes three stacked sets of "1 × 1 Conv layer and 3 × 3 Conv layer," and a 1 × 1 Conv layer. For simplicity, we name the structure as 256-128-64-32-1, where 128-64-32 represents the number of decreasing channels in order, 256 is the channel size of the input feature map, and 1 denotes the channel size of the last Conv layer, respectively.

To investigate the influence of SPB on our method, we increase or decrease the set of convolutions. Note that we do not consider SSM but preserve the FES. From Table VII, we can observe that when S4 is selected, the highest score can be obtained which gets 88.34% AP. Besides, we visualize the predicted saliency maps (see Fig. 13) of different structures. By comparing with other results, the selected SPB can obtain a more accurate saliency mask.

5) Complexity Comparision: We compare the complexity of our method with competitors in terms of FLOPs and Params. The following settings are used to evaluate all the models: 1) all the models are evaluated on a single GPU (TITAN Xp) and 2) the size of the input image is 600 × 600. Table VIII presents the results of complexity comparison, where the unit "M" denotes million and "G" denotes billion. Through comprehensively analyzing the metrics of AP (see Tables III and IV) and complexity (see Table VIII), we observe that our proposal could obtain optimal performance with relatively high computational efficiency (FLOPs) and low memory cost (Params).

6) Experiments on Using Automatically Generated Saliency Map: In this study, we use the saliency map as the extra supervision to boost the performance of the ship detector, which requires pixel-level annotation, making it time-consuming and laborious. In recent years, the studies on saliency detection are developing rapidly. Although precision is not as high as manual labeling, the salient pseudo-labels obtained by the existing saliency detection methods might be used by our proposed model to assist the ship detection task. In this regard, we conduct a series of experiments to verify the feasibility. On the HRSC-SO and DOTA-isaid-ship datasets, we first apply the classical CA [56] method to automatically acquire the saliency map, and then the generated saliency map is used to replace the salient label of the original datasets. Next, our model with the Resnet50 backbone is trained and tested on two datasets with the identical procedure introduced above. The experimental results are shown in Table IX. Although there is a slight decrease in detection performance compared with manual fine-labeling, the exploitation of the automatically generated saliency map still significantly improves the performance of the baseline. This indicates that the strategy proposed above is a solid way to strike a compromise between the time of labeling datasets and the accuracy of ship detection.

TABLE VIII
COMPLEXITY COMPARISON

Method	Backbone	FLOPs(G)	Params(M)
Faster R-CNN	Vgg16	129.39	33.63
	Resnet50	48.90	42.43
	Resnet101	74.31	63.47
Cascade R-CNN	Resnet50	109.97	68.93
	Resnet101	137.43	87.92
Libra Faster R-CNN	Resnet50	82.54	41.39
	Resnet101	110.01	60.38
SSD512	Vgg16	120.99	24.39
Mask R-CNN	Resnet50	128.47	43.77
	Resnet101	155.93	62.76
Cascade Mask R-CNN	Resnet50	249.12	77.10
HTC	Resnet50	251.71	77.23
	Resnet101	279.17	96.22
Faster ROI transformer	Resnet50	82.50	55.03
	Resnet101	110.02	74.12
ReDet	Resnet50	96.74	55.00
ANCB	Vgg16	153.88	21.24
R-DFPN	Resnet50	54.79	49.84
	Resnet101	82.21	68.83
RetinaNet	Resnet50	72.33	35.74
	Resnet101	99.76	54.73
Ours	Resnet50	78.11	38.34
	Resnet101	105.53	57.33

TABLE IX
EFFECT OF USING SALIENT PSEUDO-LABELS

Method	HRSC-SO	DOTA-isaid-ship
baseline (RetinaNet)	85.52	79.26
Ours (Automatically)	87.89	80.71
Ours (Manually)	89.10	81.03

F. Visual Results

Massive previous ship detection methods contain only two types of supervised information: the target class labels and the target bounding box labels. In our method, we use salient labels as extra supervision. To maximize the value of saliency information, our model exploits salient labels in two aspects. First, we use SSM combined with saliency information to mine more high-quality positive samples. Second, we devise a novel ship detection network with FES and SPB to further squeeze the saliency information. To validate that the two aspects indeed exploit the saliency information to promote more effective ship detection, we visualize some detection results as follows.

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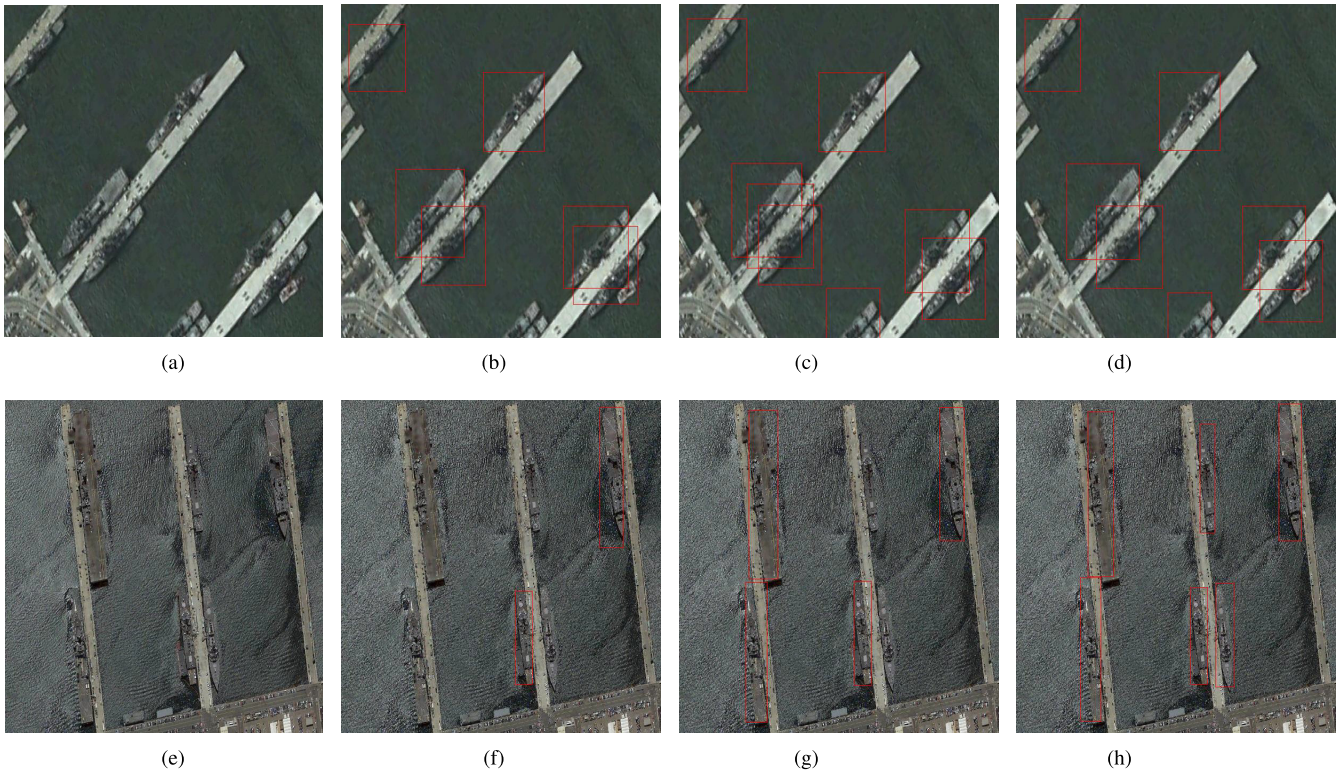


Fig. 14. Saliency information exploitation in FES and SPB. The first column is the input image and the others denote the results with different modules. (a) and (e) Image. (b) and (f) RetinaNet. (c) and (g) RetinaNet + SPB. (d) and (h) RetinaNet + SPB + FES.

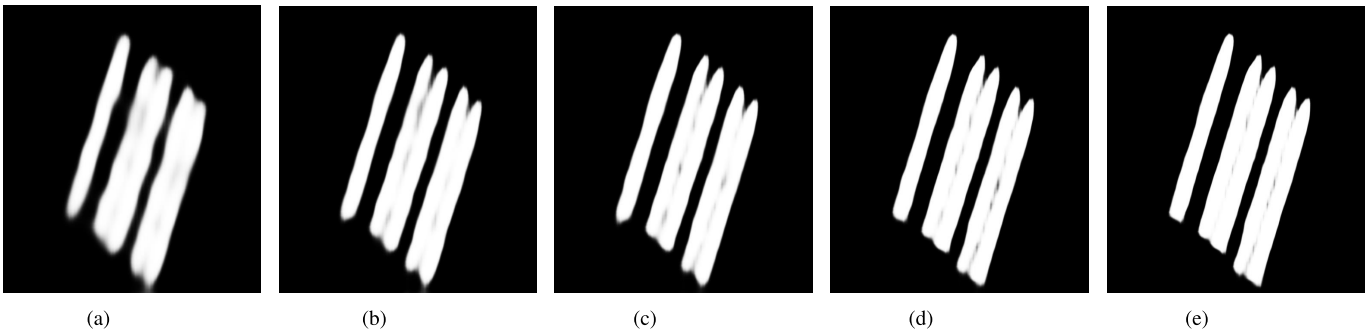


Fig. 15. Change process of the saliency mask during the training phase. The accuracy of the saliency map continuously improves as the number of iterations grows. (a) 20 iterations. (b) 60 iterations. (c) 100 iterations. (d) 300 iterations. (e) 700 iterations.

1) Saliency Information Exploitation in FES and SPB: We choose 0.8 as the confidence threshold for classification and display the results of two images in Fig. 14. It can be observed that both FES and SPB can improve the discrimination between foreground and background, which is verified by Fig. 14(f)–(h). Besides, as shown in Fig. 14(b) and (c), the network that incorporates saliency information can suppress the false alarm, such as docks. These results verify the fact that FES and SPB can obtain more discriminative information to distinguish the foreground from the background using saliency information.

2) How Does the Saliency Mask Work?: In Fig. 15, we plot the change process of saliency mask during the training phase. It can be seen that the saliency mask primarily serves as a guide in the early stages, with no obvious effect in the later training stages.

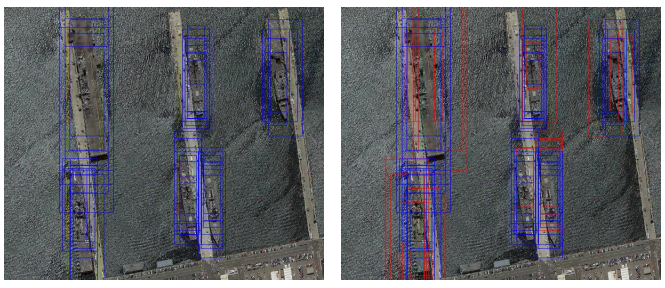


Fig. 16. Intuitive effectiveness of SSM. (a) Positive anchor boxes in RetinaNet. (b) Positive anchor boxes in SSM.

3) Saliency Information Exploitation in SSM: In HRSC-SO, we visualize the positive samples obtained by RetinaNet and

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TABLE X
MORE DETAILS OF HRSC-SO

Dataset	Train(S_{16})	Test(S_{16})	Eval(S_{16})	Total
Trainval(S_{SO})	0/436	429/453	0/181	429
Test(S_{SO})	436/436	9/453	181/181	626

RetinaNet + SSM in Fig. 16(a) and (b), respectively. As we can see, compared with RetinaNet, SSM owns the advantage of mining more positive samples and the boxes that contain more contextual information around the ship, which helps train a better ship detector.

V. CONCLUSION

This study provides new insights into ship detection aided by saliency and we analyze how to mine and use saliency information more effectively from different perspectives. In terms of structure, we innovatively propose FES and SPB for assisting the model in obtaining the foreground objects more accurately in complex background. In addition, we put forward SSM to mine more positive samples, guiding the model to generate more tight predicted boxes (i.e., the box tightly surrounding the whole body of the object). In the experiments, the results show that our method is superior to other methods. Future research will include the following: 1) considering the time-consuming and laborious pixel-wise annotation for saliency information, we will use more advanced saliency detection methods to generate more precise salient pseudo-labels; 2) in this article, we operate saliency prediction at the final low-level layer. In the future, we will explore more effective multi-head supervision strategy; and 3) it would be a promising research direction to carry out research on oriented object detection.

APPENDIX
MORE DETAILS OF HRSC-SO

We get the original data division from the ImageSets folder in HRSC2016, which includes a training set, a test set, and a validation set containing 436 images, 453 images, and 181 images, respectively. Meanwhile, as illustrated in Section IV-A, we select 429 images from the Segmentations folder in the HRSC2016 Test folder and convert them into salient labels as our trainval set, which results in data intersections between the subset of HRSC-SO and HRSC2016. Therefore, we cannot use the original data division. To describe this phenomenon, we count the overlap of data in each subset between the two datasets as shown in Table X, where S_{SO}, S_{16}, Train, Test, and Eval stand for HRSC-SO, HRSC2016, the training set, the test set, and the eval set, respectively. For the trainval set, 429 images are from the HRSC2016 test set which includes 453 images. Meanwhile, 436 images, nine images, and 181 images of our test set are from the training set, the test set, and the eval set in HRSC2016, respectively.

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Zhida Ren received the B.E. degree from the School of Electrical Engineering and Automation, Hefei University of Technology, Hefei, Anhui, China, in 2017. He is currently pursuing the Ph.D. degree with the Institute of Automation, Chinese Academy of Sciences (CAS), Beijing, China, and the School of Artificial Intelligence, University of Chinese Academy of Sciences, Beijing.

His research interests include machine learning, pattern recognition, deep learning, computer vision, and object detection.



Yongqiang Tang received the B.S. degree from the Department of Automation, Central South University, Changsha, Hunan, China, in 2014, and the Ph.D. degree from the Institute of Automation, Chinese Academy of Sciences (CAS), Beijing, China, in 2019.

He is currently an Associate Professor with the Research Center of Precision Sensing and Control, Institute of Automation, CAS. He has published several papers in major international journals and conferences, including the IEEE TRANSACTIONS ON IMAGE PROCESSING (TIP), IEEE TRANSACTIONS ON MULTIMEDIA (TMM), IEEE TRANSACTIONS ON CYBERNETICS (TCYB), IEEE TRANSACTIONS ON KNOWLEDGE AND DATA ENGINEERING (TKDE), IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS), IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY (TCSVT), *Neural Processing Letters* (NEPL), *Neurocomputing*, *Applied Intelligence* (APIN), IEEE GEOSCIENCE AND REMOTE SENSING LETTERS (GRSL), and International Conference on Pattern Recognition (ICPR). His research interests include machine learning, computer vision, and data mining.

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Zewen He received the B.S. degree from the School of Mathematical Sciences, Beihang University, Beijing, China, in 2014, and the Ph.D. degree from the Institute of Automation, Chinese Academy of Sciences (CAS), Beijing, and the School of Artificial Intelligence, University of Chinese Academy of Sciences, Beijing, in 2021.

His research interests include machine learning and computer vision.



Lei Tian received the B.S. degree from the Department of Precision Instrument, Tsinghua University, Beijing, China, in 2017. He is currently pursuing the Ph.D. degree with the Institute of Automation, Chinese Academy of Sciences (CAS), Beijing, and the School of Artificial Intelligence, University of Chinese Academy of Sciences, Beijing.

His research interests include machine learning, pattern recognition, computer vision, transfer learning, and domain adaptation.



Yang Yang received the master's degree from the Department of Astronautics, Beihang University, Beijing, China, in 2012, and the Ph.D. degree in pattern recognition and intelligent systems from the Institute of Automation, Chinese Academy of Sciences (CAS), Beijing, in 2015.

He is currently an Associate Professor with the Research Center of Precision Sensing and Control, Institute of Automation, CAS. His research interests include computer vision and pattern recognition.



Wensheng Zhang received the Ph.D. degree in pattern recognition and intelligent systems from the Institute of Automation, Chinese Academy of Sciences (CAS), Beijing, China, in 2000.

He joined the Institute of Software, CAS, in 2001. He is currently a Professor of machine learning and data mining and the Director of the Research and Development Department, Institute of Automation, CAS. He is also with the University of Chinese Academy of Sciences, Beijing. His research interests include computer vision, pattern recognition, and artificial intelligence.

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